

Retrieve place value and rounding

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Order 6,908,210, 6,890,201 and 6,980,120, then round each to the nearest 100,000. Copy or complete the first step.

2. Find the interval from -18 to 27.

3. Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.

4. Miro says: Miro applies one rounding rule to every requested place. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Order 6,908,210, 6,890,201 and 6,980,120, then round each to the nearest 100,000. Copy or complete the first step.
Answer Order: 6,890,201; 6,908,210; 6,980,120. Rounded: 6,900,000; 6,900,000; 7,000,000.
2. Find the interval from -18 to 27.
Answer 45.
3. Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.
Answer 3,550,000 to 3,649,999.
4. Miro says: Miro applies one rounding rule to every requested place. Is this correct? Explain.
Answer Mark each rounding place separately and reconsider the deciding digit each time.

Retrieve place value and rounding

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Order 6,908,210, 6,890,201 and 6,980,120, then round each to the nearest 100,000.

6. Check this result using an inverse, estimate or known fact: Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.

2. Find the interval from -18 to 27.

7. Create a seven-digit number that rounds down to 5,000,000 to the nearest million and up to 5,400,000 to the nearest 100,000.

3. Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.

8. Identify and correct this misconception: Miro applies one rounding rule to every requested place.

4. Solve this independently and show every stage: Order 6,908,210, 6,890,201 and 6,980,120, then round each to the nearest 100,000.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find the interval from -18 to 27.

10. Write and solve a similar question to: Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.

Teacher answers and checking prompts

1. Order 6,908,210, 6,890,201 and 6,980,120, then round each to the nearest 100,000.
Answer Order: 6,890,201; 6,908,210; 6,980,120. Rounded: 6,900,000; 6,900,000; 7,000,000.
2. Find the interval from -18 to 27.
Answer 45.
3. Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.
Answer 3,550,000 to 3,649,999.
4. Solve this independently and show every stage: Order 6,908,210, 6,890,201 and 6,980,120, then round each to the nearest 100,000.
Answer Order: 6,890,201; 6,908,210; 6,980,120. Rounded: 6,900,000; 6,900,000; 7,000,000.
5. Use a second method or representation for: Find the interval from -18 to 27.
Answer Expected result: 45. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.
Answer Expected result: 3,550,000 to 3,649,999. A valid check must be shown.
7. Create a seven-digit number that rounds down to 5,000,000 to the nearest million and up to 5,400,000 to the nearest 100,000.
Answer Answers vary, for example 5,360,000.
8. Identify and correct this misconception: Miro applies one rounding rule to every requested place.
Answer Mark each rounding place separately and reconsider the deciding digit each time.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Retrieve place value and rounding

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Create a seven-digit number that rounds down to 5,000,000 to the nearest million and up to 5,400,000 to the nearest 100,000.

6. Create a new example that tests the same idea as: Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.

2. Prove the answer to this question, rather than only calculating it: Order 6,908,210, 6,890,201 and 6,980,120, then round each to the nearest 100,000.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find the interval from -18 to 27.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 3,550,000 to 3,649,999.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro applies one rounding rule to every requested place.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Create a seven-digit number that rounds down to 5,000,000 to the nearest million and up to 5,400,000 to the nearest 100,000.
Answer Answers vary, for example 5,360,000.
2. Prove the answer to this question, rather than only calculating it: Order 6,908,210, 6,890,201 and 6,980,120, then round each to the nearest 100,000.
Answer Expected result: Order: 6,890,201; 6,908,210; 6,980,120. Rounded: 6,900,000; 6,900,000; 7,000,000. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find the interval from -18 to 27.
Answer Expected result: 45. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 3,550,000 to 3,649,999.
Answer One valid reconstruction is: Give the range of whole numbers that round to 3,600,000 to the nearest 100,000. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro applies one rounding rule to every requested place.
Answer Mark each rounding place separately and reconsider the deciding digit each time.
6. Create a new example that tests the same idea as: Give the range of whole numbers that round to 3,600,000 to the nearest 100,000.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.